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A system dynamics model for analyzing the eco-agriculture system with policy recommendations

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ABSTRACT

Ecological agriculture (eco-agriculture) is an approach to agriculture that seeks a balance between ecological and economic benefits to promote the sustainable development of both. This paper proposes a scientific method for analyzing the environmental and economic effects of eco-agriculture and simulating their long-term trend. Here, we focus on the eco-agriculture system of Kongtong District, Pingliang City, Gansu Province, China, and we build a system dynamics model named "AEP-SD" to evaluate the integrated effects of the system from 2009 to 2050. Under business as usual conditions, simulation results show rapid improvement until a peak is reached in 2027, after which the system will decline gradually. The model identifies some defects and disadvantages of the current agriculture system, such as the excessive increase of cattle slaughter, unstable production of methane, slow development of organic agriculture, and unsustainable energy structure. System improvement policies are offered and then proven by the model that they can indeed reduce the negative effects and eliminate the potential risks of system decline.

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1. Introduction

As the scale of human economic activity increases its presence on the globe, an ecological economic approach has arisen to account for these interactions (Gale, 2000; Ropke, 2005). As one economic activity, agriculture has the most direct and close interaction with the environment. Agricultural development is not only the basis of human survival, but also directly affects the global environment. Improving agricultural development, establishing eco-agriculture systems, and achieving good ecological and economic benefits are crucial to human development.

In recent years, eco-agriculture has been widely studied (Kleinman et al., 1995). Some studies have revealed the implication and prospect of eco-agriculture from a theoretical point of view (Altieri and Anderson, 1986; Yunlong and Smit, 1994), and some research has used case studies to demonstrate advantageous development policies of eco-agriculture (Larsson and Granstedt, 2010; Maurer, 1989; Schroll, 1994). More and more studies have taken the ecological effects into account besides economic benefits in agriculture development, and many indicators have been developed to provide decision makers with useful information, such as nitrogen use efficiency and cumulative energy (Granovskii et al., 2007; Hau and Bakshi, 2004; Hoang and Alauddin, 2011; Libralato et al., 2006; Sciubba, 2003), to assess the environmental and ecological performance of agricultural production at many scales from farms and industries to nations and the global biosphere (Hezri and Dovers, 2006; Hoang, 2011; Niemeijer, 2002; Piorr, 2003; Smith et al., 1999).

It is now evident that ecological agriculture is a complex system involving ecology, economics, industry, human behavior, policy and many other factors. A systems perspective can be used to analyze comprehensively each relevant factor of eco-agricultural development (Chen et al., 2009).

However, often eco-agriculture studies focus more on the analysis of some external influencing factors (such as the income change and the soil fertility, etc.) (Shi and Gill, 2005), and less on the industrial chain and the material-energy flow in the eco-agriculture system. The core of an eco-agriculture system is the process of material-energy production and consumption, which generates all ecological and economic effects caused by the processes. If the material-energy production and consumption cannot continuously develop, then the eco-agriculture system will decline. Therefore, to fundamentally enhance the sustainable development capacity of an eco-agriculture system, the integrated simulation and analysis of the material-energy flow processes and the trends of the ecological and economic positive-negative effects should be addressed.

Therefore, taking the case of Kongtong District, Pingliang City, Gansu Province, China, we build a system dynamics model of the eco-agriculture system named "AEP-SD"

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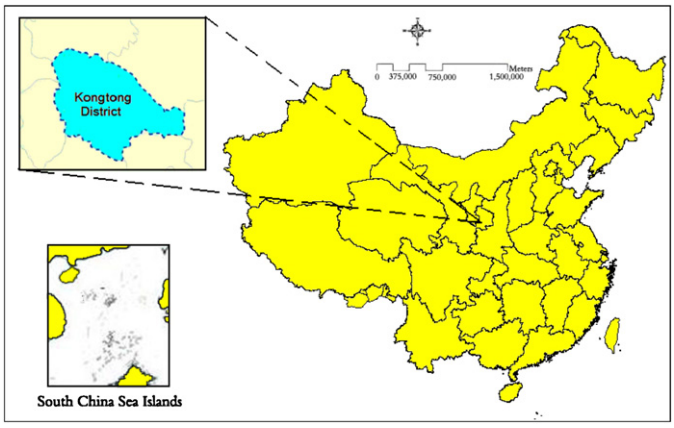


Fig. 1. The location map of Kongtong District in Pingliang City, Gansu Province, China.

(Agriculture-Effect-Policy-System Dynamics), to simulate quantitatively the material and energy flow in the local eco-agricultural industry chain, analyze the ecological economy effects and their long-term evolution trends, identify the defects of the system and then make recommendations to improve system performance. This study has important theoretical and practical values in seeking the sustainable development mode of regional ecological economy system, and more importantly the "AEP-SD" model and approach can provide a basis for similar ecological economic modeling.

2. Study area

Kongtong District (106°25′–107°21′E, 35°12′–35°45′N), Pingliang City, Gansu Province, China (Fig. 1) is located on the eastern foot of Liupan Mountain and the upstream of Jing River. Kongtong District has a semi-arid and semi-humid continental monsoon climate; the annual average sunshine duration is 2425 h; the annual solar radiation is 129.20 kcal per square centimeter; the annual average temperature is 8.6 °C; the frost-free period is 165 days; the annual average rainfall is 511 mm. Kongtong District is the hilly area on the Loess Plateau, and has serious soil erosion and fragile ecological condition. Due to the severe restriction of resource and environment, as well as the low living standard, Kongtong District is faced with the dual pressures of economy development and environment protection.

Since 2003, Kongtong District has taken the "Red bull", household production of methane, organic fruit and vegetable, and papermaking by straw as the main bodies in economic development, then eventually formed a development mode of "cattle-methane-fruit and vegetable-straw recycling". After 7-year development, by 2009, the beef cattle feeding number reached more than 300,000 head; the methane users reached 20,000 households; the output of fruit and vegetable reached respectively 69,466 tons and 250,732 tons. Comparing with the traditional development mode, in 2009, the coal burning was reduced by 15597.54 tons SCE[1]; the CO_2 emissions was reduced by 2,164,400 tons; the use amount of N fertilize, P fertilizer and K fertilizer were respectively reduced by 99,600 tons, 53,100 tons and 33,200 tons due to the utilization of organic fertilizer in farmland; the straw recycling reached 180,900 tons. Therefore, good economic and ecological benefits were achieved. However, there are still a lot of negative effects, such as the low straw recycling ratio, the CDU[2] (cattle dung and urine) pollution and the emissions of methane utilized incompletely. Especially the large-scale slaughter of beef cattle, the instability of methane production and other key problems in recent years may cause resource depletion and serious secondary pollution in the future, making the system face with the potential risk of unsustainable development.

3. Method and model description

3.1. Objectives and requirements of modeling

The ecological agriculture system brings good benefits. However, there exist some negative effects and potential risks. Therefore, it is urgently needed to build a systemic analysis model to analyze the reasons for the risks and negative effects, identify the controlling and influencing factors and then make the improvement policies for reducing the negative effects, enhancing the positive effects and promoting the sustainable development of the system.

To realize the objectives, it is required that the model built can dynamically and quantitatively simulate the development trend of the system; can reflect the interaction between the industrial development mode and the integrate effects; can reveal the key influencing factors for making the improvement policies and can test the improvement effects to insure the feasibility and effectiveness of the improvement policies.

3.2. System dynamics method

According to the above objectives and requirements, we use system dynamics method to build an eco-agricultural systemic analysis model. The system dynamics method was created by Professor Forrester of Massachusetts Institute of Technology in the mid-1950s (Forrester, 1958). After decades of development and improvement, the systemic dynamics model has been widely used in the study of economy, society, ecology and many complex systems (Chang et al., 2008; Wang and Zhang, 2001). The systemic dynamics model can reveal the dynamic changes, feedback, delay and other processes of a system, and it is characterized by quantifiability and controllability. Therefore, it has a distinct advantage in analyzing, improving and managing the system characterized by long development cycle and complex feedback effects (Tao, 2010). Therefore, the systemic dynamics method meets the modeling requirement in our study.

3.3. Logical framework of modeling

The eco-agriculture system in Kongtong District is composed of three subsystems: agriculture, effect and policy. Agriculture subsystem is mainly composed of the beef cattle feeding, the methane production and utilization, and the planting of crop, fruit and vegetable. This forms a circular industry chain with "beef cattle-methane-crop, fruit and vegetable-straw-feed (or paper)-beef cattle". The operation of agriculture subsystem generates some positive effects (such as economic growth, and yield increase) and negative effects (such as resource consumption, pollution, and greenhouse gas emissions). These effects constitute the "effect subsystem", and it can counteract on the "agriculture subsystem" by influencing ecology, economy, society and other factors. For example, the serious pollution may cause ecological degradation and even restrict agriculture development.

In order to increase the positive effects and reduce the negative effects, the decision-makers will make improvement policies according to the interaction between "effect subsystem" and "agriculture subsystem". These policies constitute the "policy subsystem". It can adjust the "effect subsystem" and eventually

[1] SCE, standard coal equivalent.
[2] CDU, cattle dung and urine.

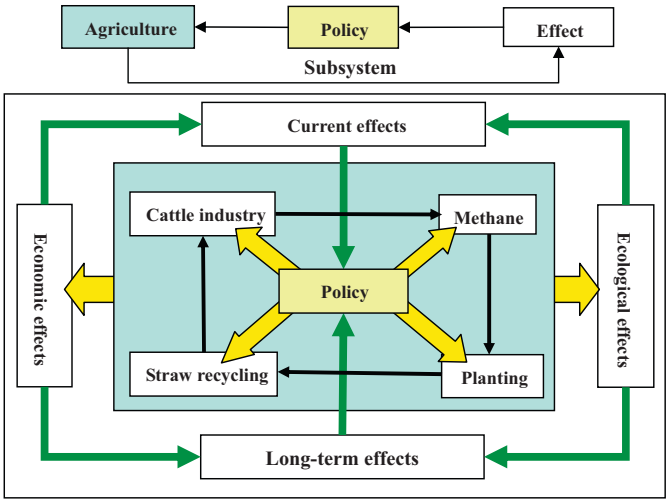


Fig. 2. Logical framework of AEP-SD model.

promote the sustainable development of the whole eco-agriculture system (Fig. 2).

According to the interaction among agriculture, effect and policy, we build a system dynamics model named "AEP-SD" (Agriculture-Effect-Policy-System Dynamics).

3.4. Data sources

The data used in this model mainly come from the first-hand information from field investigation, the results of the questionnaires and face-to-face interviews in local areas, "2000–2009 Statistical Yearbook in Kongtong District and Pingliang City", and "2000–2009 Economic Statistics Report in rural areas of Kongtong District". The conversion coefficients of energy in the model mainly base on the standard of "Chinese Energy Statistics Yearbook", and the carbon emissions factors of straw and methane burning base on "2006 IPCC Guidelines for National Greenhouse Gas Inventories". The calorific value and carbon emissions coefficient of coal are respectively 4933 kcal/kg and 1.9779 according to the test results provided by the local power plant.

4. The modeling process of AEP-SD model

4.1. Causal loop diagram

According to the logical framework and the material-energy flow in agricultural industry chain, the causal loop diagram of the AEP-SD model is designed. In the diagram, the blue arrows show the generating paths of some positive effects such as eco-agriculture output, CO_2 emissions reduction and so on; the green arrows show the generating paths of some negative effects such as the resource consumption, pollution emissions and so on; the red arrows show the paths of formulation and implementation of system improvement policies (Fig. 3).

4.2. Stock-flow diagram

Stock-flow diagram is the core of AEP-SD model, and is the process of quantization and materialization of causal loop diagram. On the basis of the actual data about eco-agriculture system from 2003 to 2008 and the differential equations built by stock-flow diagram, the whole eco-agriculture system is simulated quantitatively and dynamically (Fig. 4) (all the functions and parameters are shown respectively in Appendices A and B (Vensim software formats)). To facilitate discussion, the model is divided to four parts according to the industry chain.

4.2.1. Part 1. Feeding and slaughter of beef cattle

This part reflects the changes in cattle feeding, cattle slaughter and other relevant factors. This part of model includes 3 level variables[3]: "calf", "mature cattle" and "marketing cattle". These variables are controlled by 4 rate variables[4]: "breeding rate", "fattening rate", "marketing rate" and "slaughtering rate", and influenced by other 22 auxiliary variables (Fig. 4.1).

In the cattle feeding chain, the mature cattle are marketed and slaughtered after calf feeding, fattening and growing up. "Slaughter capacity" and "cattle marketing number" determine the relation between the supply and demand for beef cattle, and thereby affect the purchase price of beef cattle and "feeding profit". The profits affect the farmer's initiative in increasing the beef cattle number and determine "the breeding rate" and the "calf" number. These processes form a negative feedback loop. In addition, "the total number of beef cattle" determines the demand for straw feed and affects the development of straw feed industry upstream. The "CDR" amount affects the methane production downstream (Fig. 4.1).

The cattle feeding industries in Kongtong District develop rapidly in recent years. Due to the governmental financial support and tax break to cattle feeding households and slaughter enterprises, the breeding rate and slaughter number of beef cattle rise continuously, However, the slaughter enterprises are few and small-scale, and the beef cattle are slaughtered by farmers themselves, therefore, "the slaughter rate" is lower than "the breeding rate", and the total number of beef cattle rise rapidly.

In 2009, the total number of beef cattle reached 310,084 head; the profit of cattle feeding reached CNY[5] 182.6 million; the CDR reached 2,945,800 tons and was mainly utilized in methane producing and farmland fertilizing; the by-product of cattle bone reached 14506.8 tons and the cattle blood and viscera reached 38684.8 tons. However, due to the bad sanitary condition and farmers' low-tech slaughter way, these by-products cannot meet the requirement of bio-pharmacy and food. Therefore, they cannot be utilized effectively and even are thrown as garbage, leading to the huge resource waste and ecological pollution, which become the negative effect of the eco-agriculture system.

4.2.2. Part 2. Production and utilization of methane

This part reflects the production and utilization of methane, the utilization of organic fertilizer and the development of organic fruit and vegetable, and it is the main unit of positive effects generation in eco-agriculture system. This part of model includes 3 level variables: "the methane stock", "the stock of MSR[6]" (methane slurry and residue) and "the income increment of organic fruit and vegetable", 9 rate variables such as "the methane production rate" and 25 auxiliary variables (Fig. 4.2).

In this model, "the proportion of CDR used in methane production" determines "the methane producing rate". "Methane utilization rate" reflects the annual utilization amount. In 2009, the methane production and consumption reached respectively 7,084,640 m^3 and 4,959,250 m^3, making the burning coal reduce by 15531.4 tons. The good ecological and economic benefits are achieved. However, due to the diurnal and seasonal temperature

[3] Level variables: expresses the stock amount of variables.
[4] Rate variables (** rate): expresses the annual variation values of variables. For example, "slaughtering rate" expresses the annual slaughter amount of beef cattle.
[5] CNY, the unit of Chinese currency, 1 Yuan = 0.153808 dollar (the exchange rate on 28th, April, 2011).
[6] MSR, methane slurry and residue.

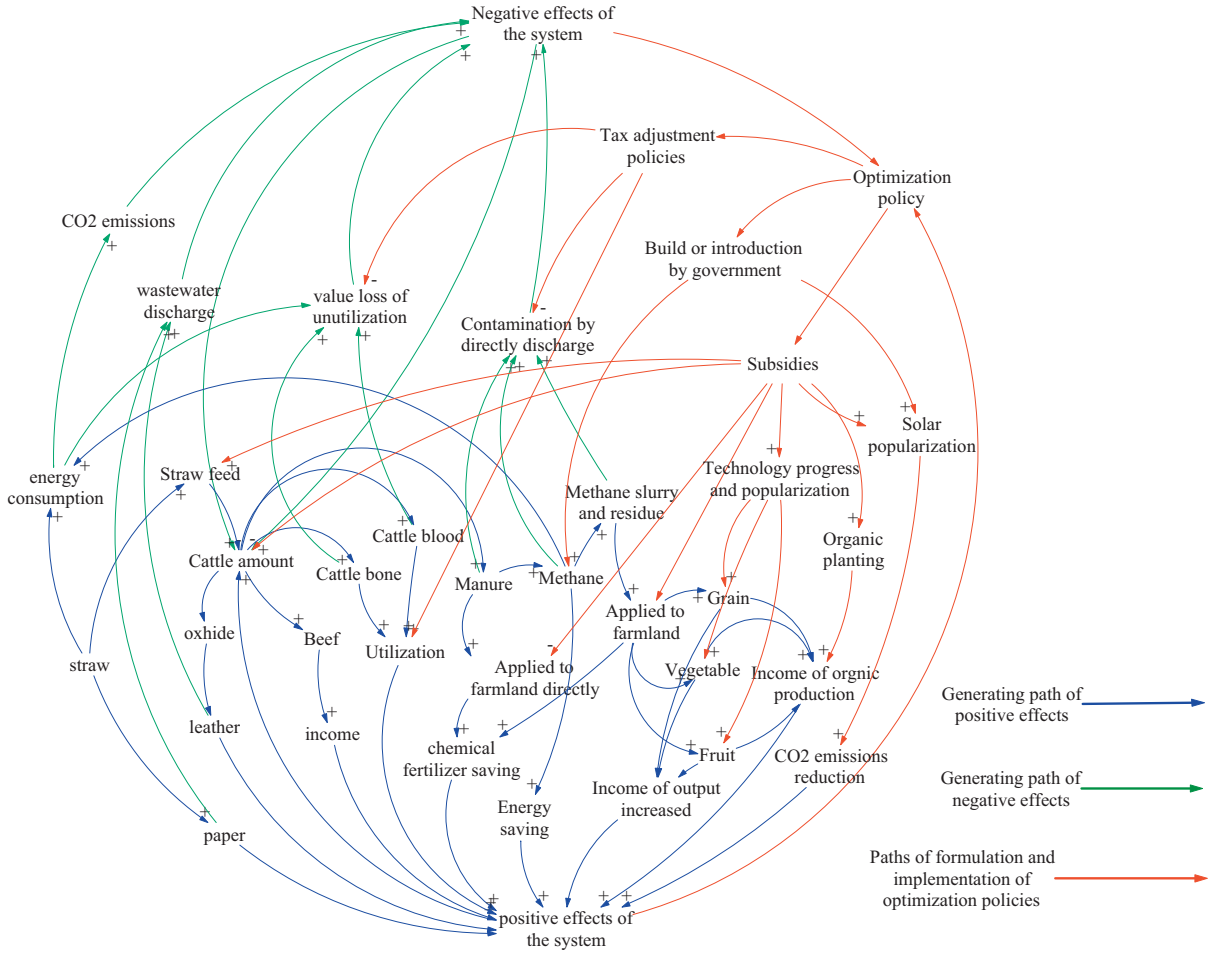


Fig. 3. The causal loop diagram of AEP-SD model.

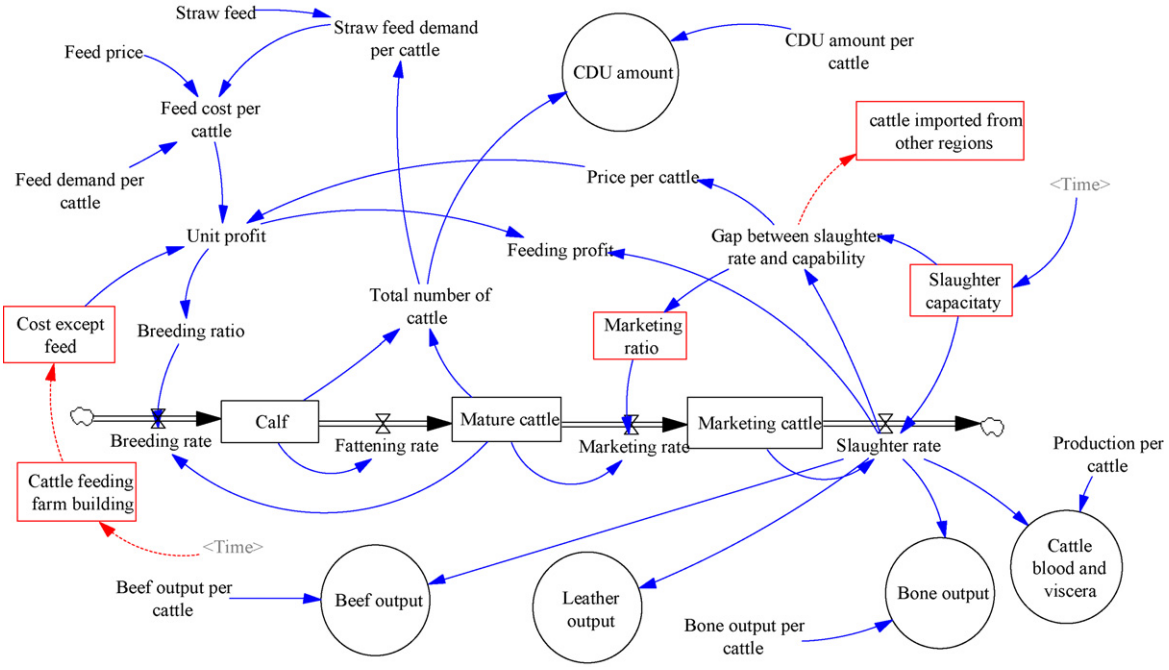


Fig. 4.1. The stock-flow diagram of beef cattle feeding and slaughter.

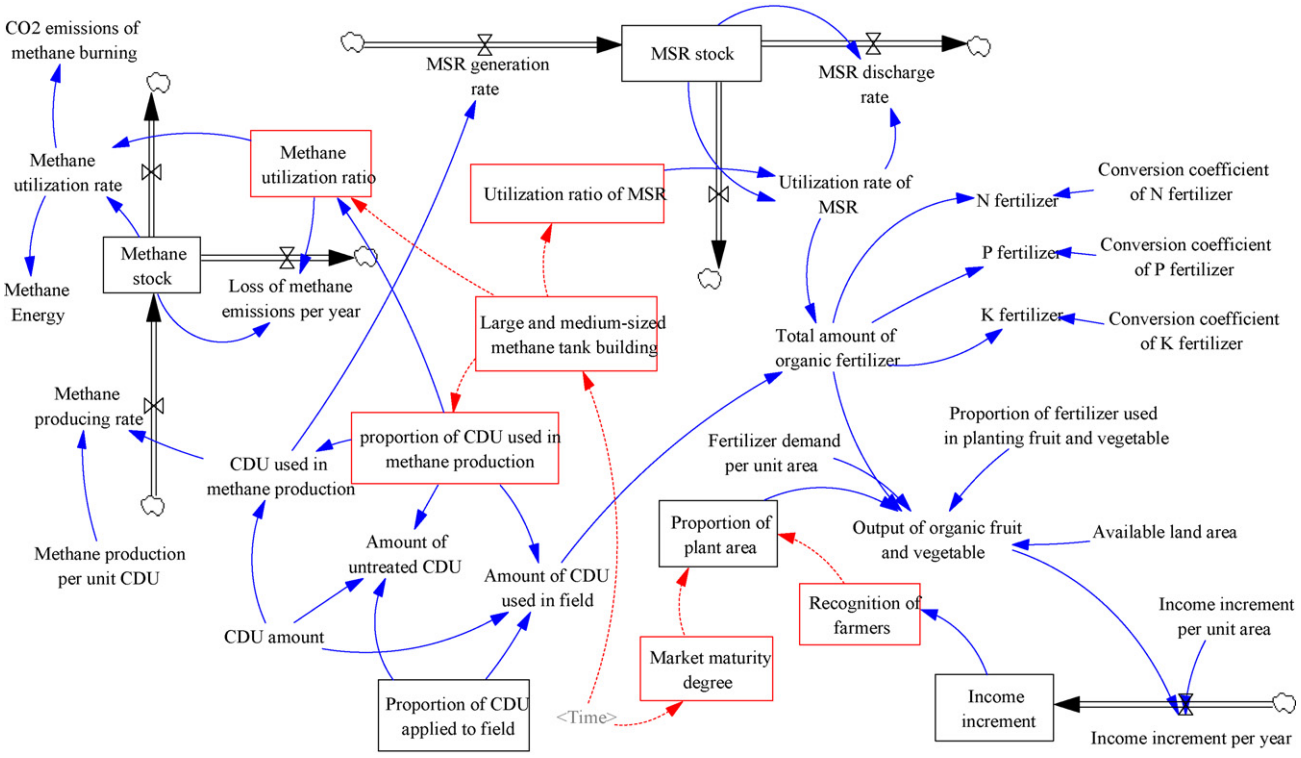


Fig. 4.2. The stock-flow diagram of methane production and utilization.

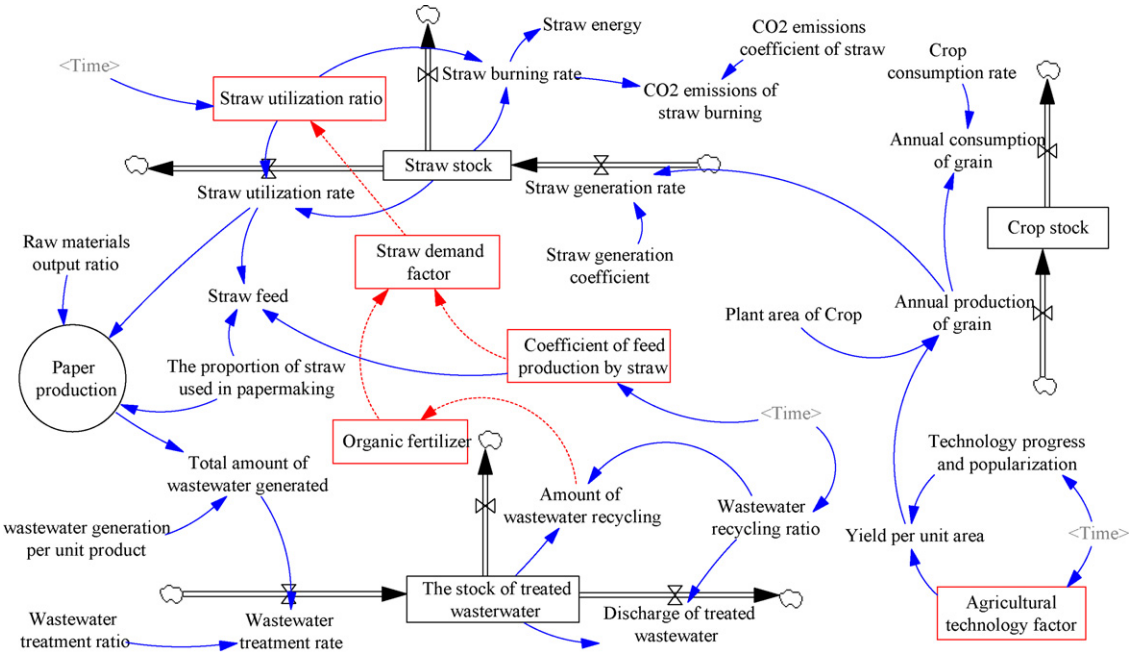


Fig. 4.3. The stock-flow diagram of crop production and straw recycling.

differences, as well as the low-tech equipment in household production, the methane production amount and concentration are very unstable, and each year about 30% methane cannot be utilized because of the leakage or not meeting the use requirement. The emissions of methane unutilized leads to waste and pollution, and expressed by the variable of "loss of methane emissions per year" in the model.

The CDU unutilized in methane production and the MSR generated by methane production are mainly used in farmland as organic fertilizers besides discharged with a small part. In 2009, the "total amount of organic fertilizer" is equivalent to 99,600 tons N fertilizer, 53,100 tons P fertilizer and 33,200 tons K fertilizer, replacing much utilization of chemical fertilizer and providing the basis for the development of organic fruit and vegetable.

However, the development of organic fruit and vegetable is restricted by "farmers' recognition" and "the perfection degree of market". By field investigation, we finds that 80% farmers do not know the market demand for organic fruit and vegetable, and the transaction channel of organic fruit and vegetable are not formed, making the planting area still be little by now (Fig. 4.2).

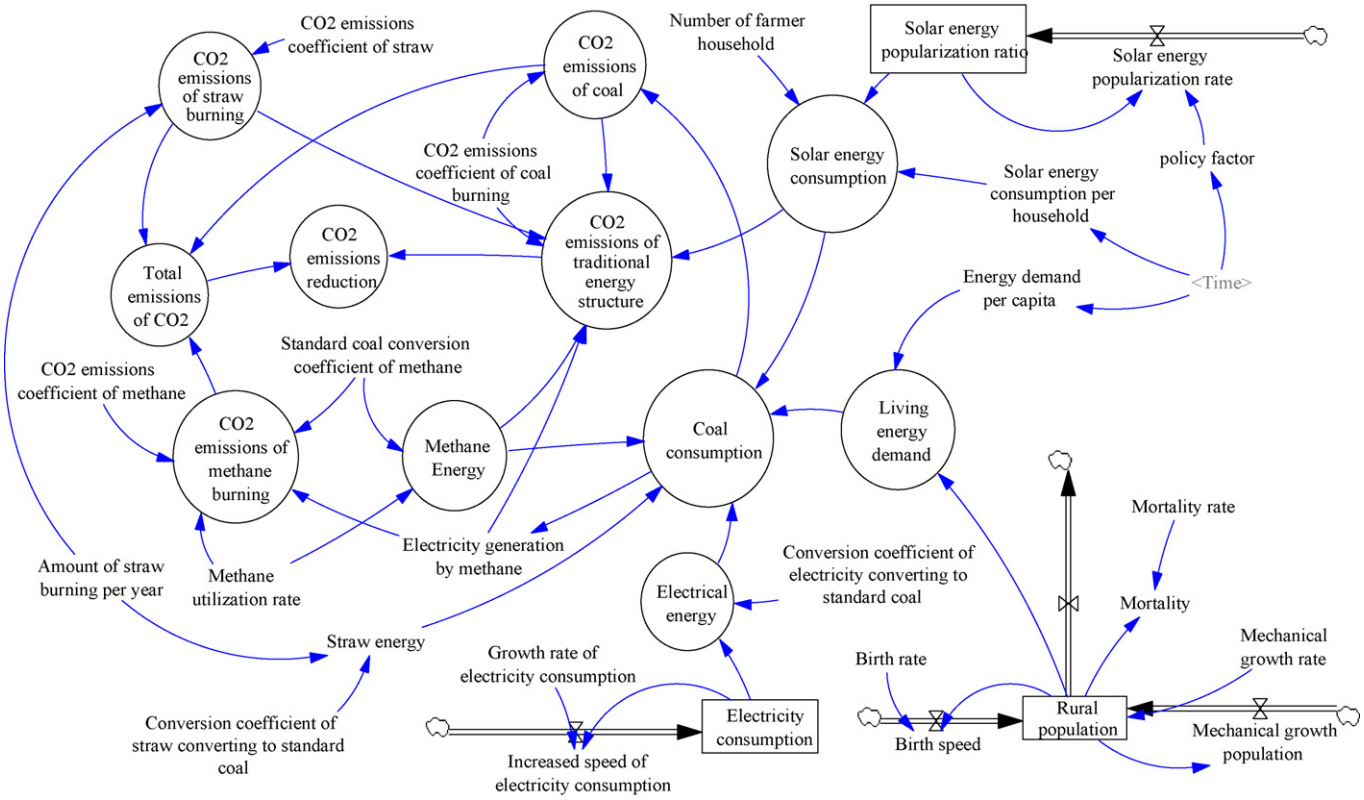


Fig. 4.4. The stock-flow diagram of energy structure and carbon emissions.

4.2.3. Part 3. Crop production and straw recycling

This part reflects the crop production and straw recycling. The part of model includes 3 level variables: "the straw stock", "the feed stock" and "the treated wastewater stock", 8 rate variables such as "the straw utilization rate", and 24 auxiliary variables (Fig. 4.3).

Hypothesizing the area of crop sowing is constant, the popularization and progress of technology are the main driving force for the yield increase of crop and straw. The straw in Kongtong District are recycled for papermaking and feed production. "The utilization rate" determines the annual production amount of paper and feed. A large amount of wastewater generated by papermaking is reused after treatment, and a small part discharges or evaporates. High cost of wastewater treatment restricts the purchase price of straw and thereby the straw utilization rate. The straw unutilized is burned as fuels by farmers and brings large CO_2 emission.

4.2.4. Part 4. Energy structure and carbon emissions

This part reflects the demand and consumption of different kinds of living energy, as well as the CO_2 emissions in the rural areas in Kongtong District. This part of model includes 3 level variables: "the total rural population", "the utilization amount of electric energy" and "the popularization ratio of solar energy", 5 rate variables and 28 auxiliary variables (Fig. 4.4).

The population quantity and the energy demand per capita in Kongtong District determine the total energy demand. Before applying the eco-agriculture mode, the energy types in Kongtong District are mainly coal, electricity and straw. Now, by straw recycling, methane popularization and solar energy utilization, the burning of coal and straw, as well as the CO_2 emissions are reduced significantly, and the energy consumption structure is changed. Coal, electricity, methane, straw and solar energy are the main energy types currently. According to model hypothesis (5), the gap between the total non-coal energy consumption and total energy demand is the coal consumption.

In the model, the actual CO_2 emissions is the sum of CO_2 emissions caused by various fossil energies, and the gap between the actual CO_2 emissions and the CO_2 emissions caused by traditional energy such as coal, electricity and straw is the amount of CO_2 emissions reduction, which can be used to weigh the benefits of CO_2 emissions reduction brought by eco-agricultural development mode.

4.3. Hypotheses and boundaries of the model

(1) The beef cattle in the model are all the "Pingling Red Bull" produced specially in Kongtong District and surrounding areas.
(2) The feeding zones and slaughter enterprises all concentrate in Kongtong District and surrounding counties, moreover, the out-of-town beef cattle are few. Therefore, the model hypothesizes the cattle price is decided by local supply and slaughter. The hypothesis accords with local realities.
(3) The model does not consider the sudden change in the beef cattle number caused by emergencies such as large-scale animal disease.
(4) According to the farm land policies in China, when a land is occupied, a new one needs supplementing with the same area in the administrative region. Therefore, the model hypothesizes the farm land area in Kongtong District is constant
(5) Coal has the highest purchase price in local living energy, therefore, the model hypothesizes the farmers will choose a cheaper energy to replace it.

4.4. Model test and validation

Model test and validation aim at justifying the reliability of the model and providing confidence for model application.

Using the tool of "Run Reality Checks" in Vensim software, we input and run the "Reality Check functions" to check whether the

logic relations among variables are logical and real. The check results can be outputted automatically. For example, we input the function: "THE CONDITION: Total amount of cattle = 0, IMPLIES: Methane energy = 0" to test whether the "Methane energy" is zero under the assumed condition "Total amount of cattle" is zero. If the "Methane energy" is zero, the reality check is passed (since methane is produced by CDR, without cattle the methane energy cannot be gained). Test indicates, under the assumed condition, the "Methane energy" is indeed zero and the model is right. Using the same method, we check all the variables in the model. Results show that all the reality checks are passed.

In addition, we also make a sensitivity test of the model. We use the tool of "automatically simulate on change", although without the special tool for sensitivity test in Vensim.ple version. Through varying each parameter value from maximum to minimum, we test whether the simulation results of relevant variables accord with logic. For example, we vary the value of the parameter "proportion of CDU used in methane production" from 0 to 1 and then make simulation. When the value is equal to 0 (the CDR is completely not used in methane production), the "methane stock", the "CO_2 emissions of methane burning" and other variables related with methane are equal to 0; when the value is equal to 1 (the CDR is completely used in methane production), the "methane stock", the "CO_2 emissions of methane burning" and other variables related with methane increase by about 10 times, and the regional CO_2 emissions decreases by about 50%, according with the probable variations of system under the extreme conditions. Moreover, during the parameter variation, the simulation trends of relevant variables do not change, demonstrating the response of model to the variation of this parameter is sensitive and in accord with logic. By the same method, all the variables in the model are tested and passed the sensitivity test.

5. System analysis applying the AEP-SD model

By the model, the probable evolution trends of the eco-agriculture system between 2009 and 2050 under business as usual conditions are simulated and analyzed in Kongtong District, and the simulation process is named "current".

5.1. Beef cattle feeding and slaughter

Simulation results show, due to the support of tax breaks, the slaughter enterprises will develop rapidly, and the slaughter rate will rapidly rise and gradually exceed the cattle breeding rate in the future. By the year 2026, the total number of beef cattle will rise until a peak is reached 392,116 head, after which it will decline rapidly.

After 2050, the cattle supply will decrease; the slaughter rate is much lower than the slaughter capacity; the total number of beef cattle will be steadily less than 100,000 head, causing the corresponding decrease of CDR generated, straw feed demand, methane production downstream and straw recycling upstream. The system may decline. In addition, if the slaughter way is not changed, cattle bone, cattle blood and other by-products will still be difficult to be utilized fully in the future, resulting in huge waste and pollution (Fig. 5).

5.2. Production and utilization of methane

Simulation results show that with the variation of beef cattle number, the CDU amount increases and then decreases, making the annual methane production rise to the peak of 8,958,860 m^3 in 2026 and then gradually decrease to less than 2,500,000 m^3. In addition, if the household production way is not changed, much MSR will be difficult to be fully utilized and bring pollution due to the backward equipment, inconvenient operation and other shortcomings. Moreover, every year there will be more than 2 million m^3 methane emitting directly, resulting in huge waste and pollution and making the methane industries difficult to develop. The organic fertilizer used in farmland will also be largely reduced after 2027, making the organic agriculture difficult to develop and eventually leading to the farmland production still depending on much chemical fertilizer (Fig. 6).

5.3. Crop production and straw recycling

In the future, with the popularization and advancement of dry farming technique, crop production and straw recycling will gradually develop year by year, and the ecological and economic benefits will constantly increase. However, the annual amount of wastewater generated by papermaking will increase from 2.33 to 5.94 million tons between 2009 and 2040, and only about 70% can be reused. The high cost of wastewater treatment makes the purchase price of straw difficult to increase and restricts the straw utilization ratio. In the future the actual amount of straw burning will constantly increase in Kongtong District, and after 2039, it will be more than 50,000 tons. The annual CO_2 emissions amount will be more than 77,600 tons (Fig. 7).

5.4. Energy structure and CO_2 emissions

By developing methane and solar energy, as well as increasing the electricity consumption proportion, the coal is saved much in Kongtong District. By 2036, the non-coal energy utilization will reach 52558.46 tons SCE, and the energy cost and CO_2 emissions will be greatly reduced. By 2027, the CO_2 emissions reduction

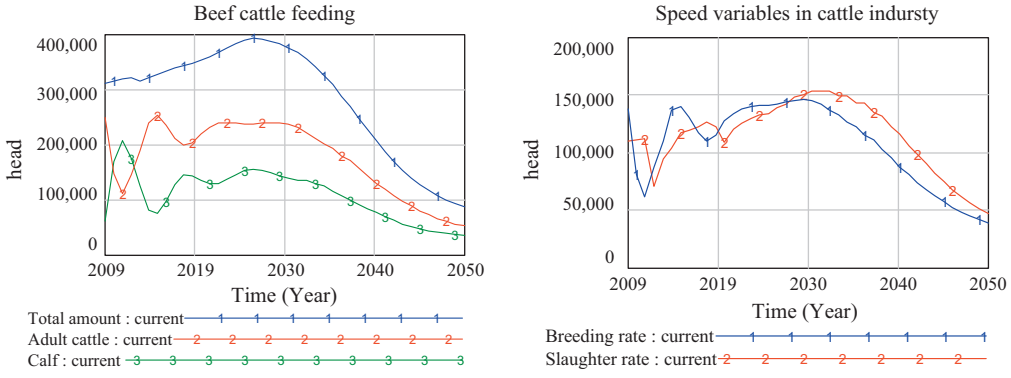


Fig. 5. Simulation of beef cattle feeding and slaughter between 2009 and 2050.

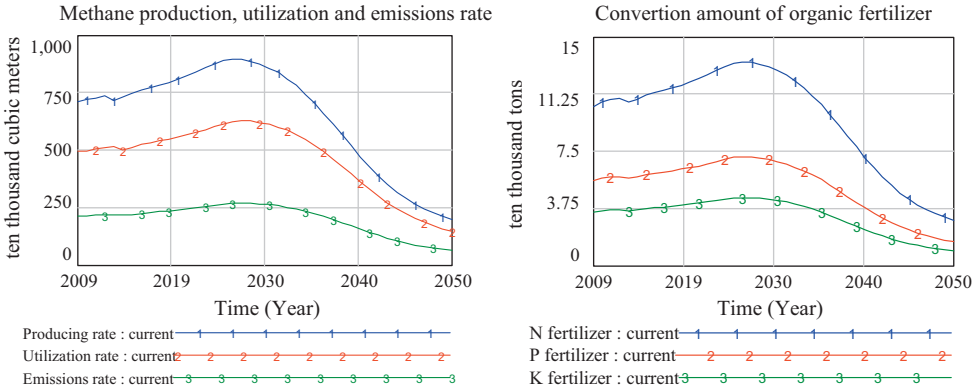


Fig. 6. Simulation of methane production and utilization between 2009 and 2050.

amount will reach 32,500 tons. However, with the decline of methane energy, the energy structure will be gradually return to coal burning, the CO_2 emissions reduction amount will gradually decrease after 2028, and the energy saving and emissions reduction will not continue (Fig. 8).

5.5. Potential risks and negative effects

Based on the predictions and analyses, some major potential risks and negative effects are identified in the eco-agriculture system in Kongtong District.

(1) Slaughter capacity increases rapidly and exceeds the breeding rate, which may lead to the sharp reduction of local beef cattle, cause the declines of methane production, organic agriculture and straw recycling, and eventually result in the unsustainable development of the system. In addition, the bad conditions of sanitation, collection and processing during the household feeding may cause the waste and discharge pollution of the by-products (such as the cattle blood).

(2) Government subsidizing the household production of methane is not conducive to the application of advanced technology and facilities, resulting in the instability of methane production, methane emissions and MSR pollution and other negative effects.

(3) Farmers have not recognized the benefits of organic agriculture and the market is immature, restricting the development of organic agriculture and bringing the negative effects such as the large utilization of chemical fertilizer, the low utilization ratio of organic fertilizer and the CDU pollution.

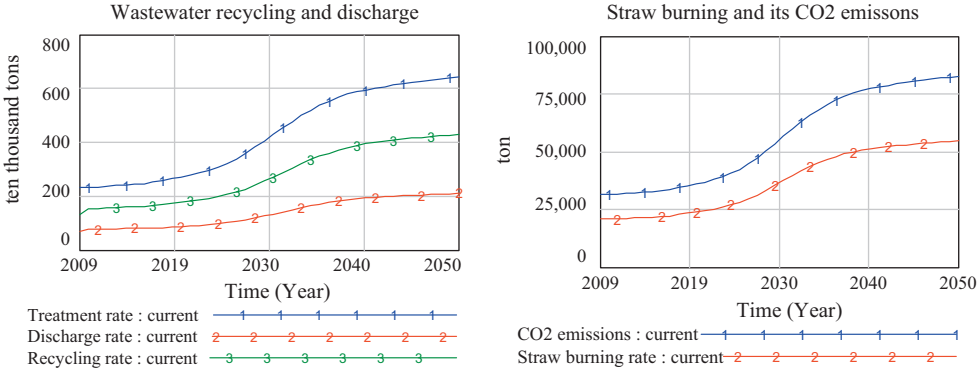


Fig. 7. Simulation of crop production and straw recycling between 2009 and 2050.

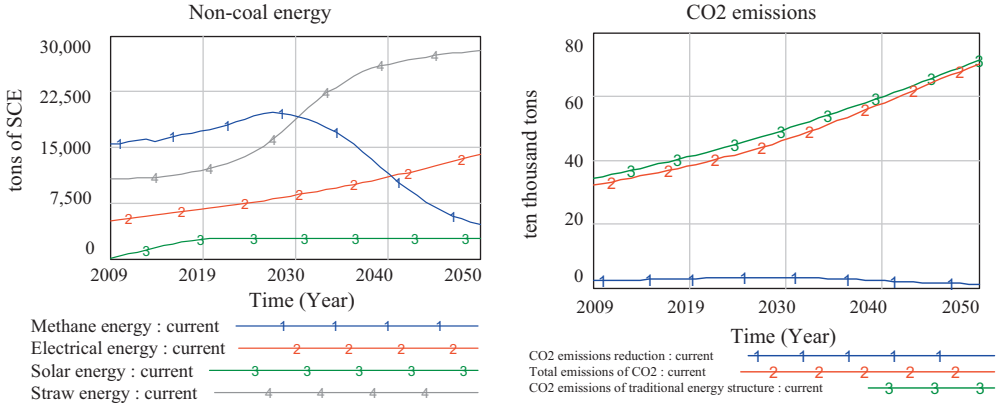


Fig. 8. Simulation of energy structure and carbon emissions between 2009 and 2050.

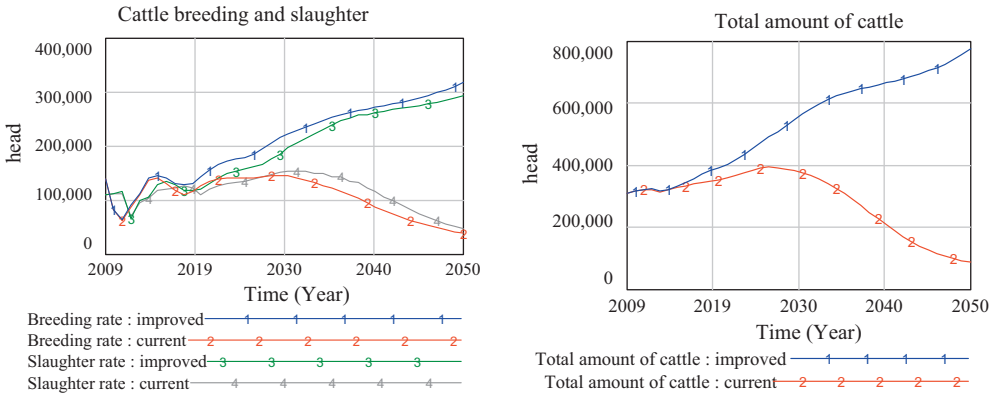


Fig. 9. Simulation of beef cattle feeding and slaughter under the condition of policy implementing.

(4) The cost of wastewater treatment is high and the recycling ratio of wastewater is low, limiting the purchase price of straw and resulting in the low recycling ratio of straw and large CO_2 emissions caused by burning.

6. Improvement policies and effects prediction

By the AEP-SD model, not only the eco-agriculture system under business as usual condition can be simulated and analyzed, but also the system improvement policies can be made based on the simulation results, then the integrated effects changes under the condition of policy implementing can be simulated (the simulation process is named "improved")to judge the effectiveness of policies and eventually the scientific and feasible adjustment projects can be determined. The system improvement policies make some variables and functions change directly and simultaneously make some new variables and material flow process increase in the model. These variations are manifested by the red arrows and red boxes in Fig. 4 and shown in the Appendix C (VEISIM formats).

6.1. Improvement policies regarding cattle feeding and slaughter

The improvement objectives are eliminating the potential risks such as the sharp reduction of local cattle species and the decline of relevant industries; eliminating the waste and discharge pollution of by-products such as the cattle blood.

6.1.1. Improvement measures

(1) Government should stop subsidizing the household feeding and turn to support the construction of 40–50 large-middle cattle farms. By 2020, more than 95% cattle in the district should be centralized in the cattle farm, and the farmers can hold the farm stocks or work in the farm; (2) government should cancel the tax breaks to slaughter enterprises until the cattle farms are finished constructing, then the tax breaks policy can be implemented for 5 years; (3) Government needs to construct the trade market and logistics center for livestock (finish it at 2015).

6.1.2. Changes in the model

The new increased variable "cattle farm building" in the model expresses the cattle farm building schedule. The increase of cattle farm makes cattle feeding more scientific and effective and thereby reduces the "cost except feed". According to the field investigation, the "cost except feed" in the cattle farm built is 5.5–10.5% lower than that in household feeding. Therefore, by 2020, the average feeding cost per cattle will decrease by about 5% at least, making the "breeding rate" increase by 11.52%; Simultaneously, the tax control policies can make the growth speed of the "slaughter capacity" slow down by 13.165% before 2020 and then moderately rise from 2020 to 2050, making the "slaughter rate" and "breeding rate" keep long-term balanced and the "total number of cattle" continuously rise. The establishments of trade markets and material flow centers may attract much "cattle imported from other regions", relieving greatly the pressure of local cattle supply (Fig. 4.1).

In addition, by the cooperation between cattle farms and slaughter enterprises, all the beef cattle can be slaughtered in the slaughterhouses, and the sanitation conditions and the processing capacity can be improved greatly, making the comprehensive application of cattle bone and blood, to the bio-pharmaceutical industries development become probable, eliminating the waste and pollution of the by-products and creating new economic value.

6.1.3. Predictions of improvement effects

Stimulation results show that, by 2050, the average feed cost will decrease by 13% and the breeding rate will rise by 129%; slaughter rate and breeding rate will keep basically balanced; the total number of cattle will steadily rise to 772,932 head and the annual slaughter number will reach more than 290,000 head by 2050, avoiding the probable bad ecological result of the sharp reduction of cattle species (Fig. 9).

6.2. Improvement policies regarding methane production and its comprehensive utilization

The improvement objectives are improving the stability of methane production and reducing the discharge pollution of methane, CDR and MSR; promoting the development of organic agriculture and reducing the utilization of chemical fertilizer.

6.2.1. Improvement measures

(1) Government should stop subsidizing the household methane tanks construction, turn to subsidize the cattle farms for centralized construction, and then lay pipelines for methane transport. By 2020, government should complete the construction of the methane supply system covering the rural areas. (2) The agricultural management should finance the construction of trade market and network sale platform for organic fruit and vegetable. (3) The specialized institutions should be founded for farmers' technological training and organic products propaganda, improving farmers' recognition level on the organic agriculture.

6.2.2. Changes in the model

The new increased variable "large and medium-sized methane tank building" in the model expresses the building speed. The advanced equipment and technology in large methane tanks construction can eliminate the seasonal temperature difference, ensure

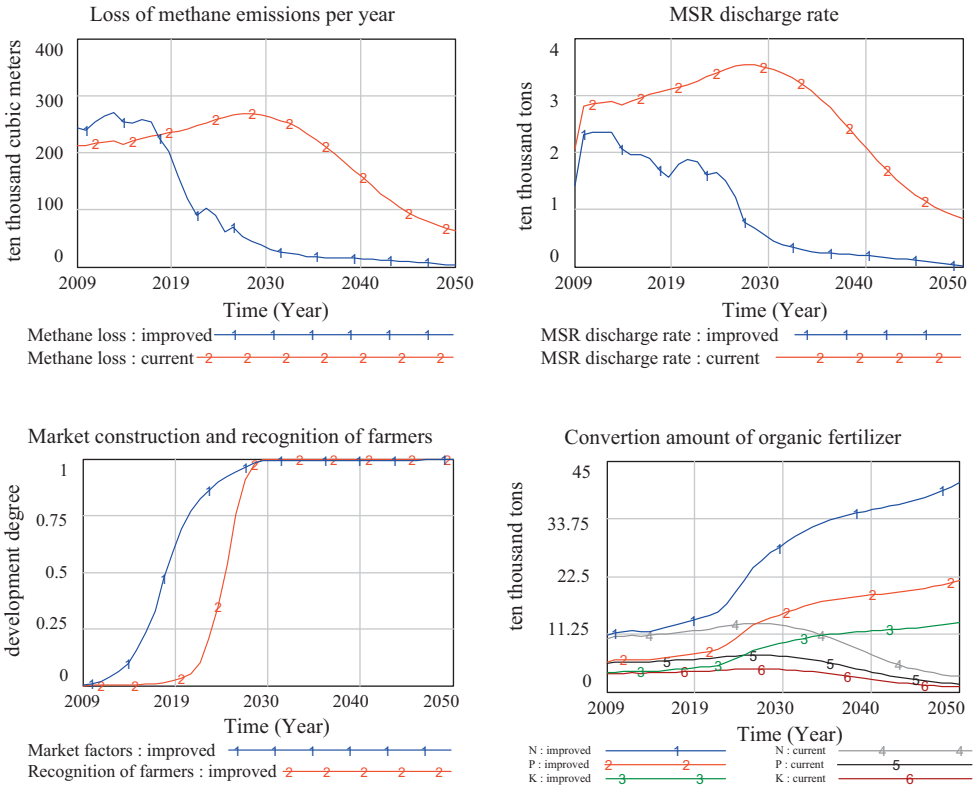


Fig. 10. Simulation of methane and organic planting under the condition of policy implementing.

the stability of methane production and supply, raise the production amount, bring convenience and security for farmers' utilization and make MSR and CDU get centralized treatment, which may make the "proportion of CDU used in methane production", "methane utilization ratio" and "utilization ratio of MSR" gradually rise to near 100%, and eliminate the discharge pollution of methane, CDU and MSR. In addition, All the CDU are converted completely into MSR, and then the comprehensive utilization of MSR not only can improve fertility, but also can be used in seed soaking and pest control, increasing the "agricultural technology factor" by 4.51–22.3% and largely promoting the yield increase of crop, fruit and vegetable. The establishment and operation of trade market and train institution may gradually improve the "market maturity degree", the "recognition of farmers" and then the "proportion of plant area" (Fig. 4.2).

6.2.3. *Predictions of improvement effects*

Simulation results show, by 2020, the annual output of methane will reach 55,385,800 m^3, equivalent to 173,457 tons SCE; after 2041, the annual output will steadily be more than 300 million m^3, equivalent to 939,540 tons SCE; by 2021, the methane loss will decrease to 851,600 m^3, about 30% of that in household production; by 2030, methane emissions pollution will be basically eliminated.

In the early development stage of organic fruit–vegetable market, farmers' recognition on organic agriculture lags behind market maturity degree. However, by 2022, when the market maturity degree reaches about 0.82,[7] the information regarding higher income achieved by organic production may rapidly spread. Then the organic agriculture may be rapidly accepted and the farmers' recognition eventually matches to the market maturity degree in 2028. By then, the planting area of organic fruit and vegetable may expand to 7598 acres; the income increment may reach 140.036 million; the organic fertilizer in the form of MSR used in farmland may reach 4,422,380 tons, about 30 times of that under business as usual conditions, and after 2035, it may reach more than 5,500,000 tons, equivalent to 330,000 tons N fertilizer, 170,000 tons P fertilizer and 110,000 tons K fertilizer, greatly reducing the utilization of chemical fertilizer in the whole District (Fig. 10).

6.3. *Improvement policies regarding energy, carbon emissions and straw recycling*

The improvement objectives are reducing the cost of wastewater treatment and improving the wastewater reuse ratio in papermaking industry; increasing the straw utilization and decreasing the straw burning; reducing regional CO_2 emissions.

6.3.1. *Improvement measures*

(1) Government should provide loan with low interest to encourage papermaking industries to introduce the technology regarding black liquid producing organic fertilizer. (2) Government should provide preference of land use to help the cattle farm to build the factory regarding the feed processing by microbial fermentation. (3) Government should popularize new energy and subsidize the methane power generation and solar power.

6.3.2. *Changes in the model*

By the loan with low interest, the papermaking industries can introduce special technology and equipment to convert the black liquor generated by papermaking into water and multi-element compound fertilizer (expressed by the variable "organic fertilizer"), not only bringing new economic benefit and reducing the

[7] Market maturity degree: "0" expresses having no market; "1" expresses the completely mature market.

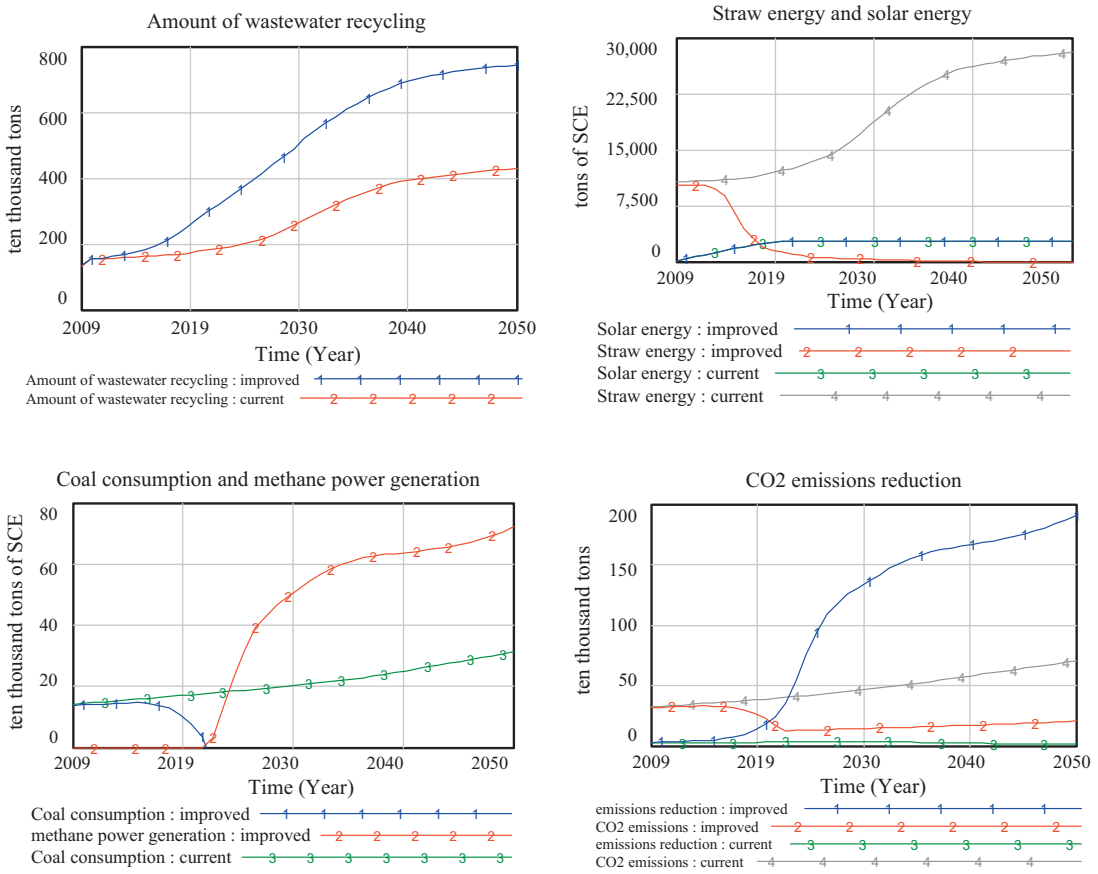


Fig. 11. Simulation of energy structure and carbon emissions under the condition of policy implementing.

wastewater treatment cost, but also increasing the "wastewater recycling ratio" to more than 99%. Simultaneously, the preference of land use promotes the build of feed processing factory, making the "coefficient of feed production by straw" increase by 12%. The implementing of these two measures may increase the straw demand (expressed by the new variable "straw demand factor"), improve "straw utilization ratio" and reduce "straw burning rate" (Fig. 4.3). In addition, since 2008, government has subsidizes CNY 150 (the total fee is CNY 160) to the household who will install the solar cooker in the test areas, and now, 2000 households have used the solar energy. In the future, the subsidies will be extended to the whole region, and the methane power generation can develop when the methane is sufficient, greatly reducing the coal utilization and CO_2 emissions. By 2020, the technology regarding black liquor producing organic fertilizer will be applied widely, making the annual income of papermaking enterprises increase by 20–30%. By 2035, the annual amount of wastewater reuse will increase to 6,200,000 tons, rising by 5 times and realizing zero discharge of wastewater. By 2028, the straw recycling ratio will reach 100%, and no straw will be burned. In addition, by 2019, the utilization of solar energy will cover 90% of all rural regions, replacing the annual coal burning of 2693 tons SCE (Fig. 4.3).

6.3.3. *Predictions of improvement effects*

The methane energy will increase continuously, and by 2030, the theoretical electricity generation will reach more than 500,000 tons SCE. Clean energy development will reduce the coal use rapidly. By 2022, theoretically all the coal can be replaced by other energies, making the CO_2 emissions reduce from 320,000 tons to 125,700 tons, and the annual average emissions reduction amount will be 17 times of that before improvement (Fig. 11).

7. Conclusions and discussion

By utilizing the AEP-SD model, the eco-agriculture system in Kongtong District is simulated, analyzed and improved, and some conclusions are drawn as follows:

(1) In this study, we build and use a system dynamic model (AEP-SD) to simulate and analyze an eco-agriculture system case. Based on the simulation results, the potential risks and negative effects of the system are identified, and then the system improvement policies are made and proven that they can indeed eliminate all the risks, reduce and negative effects and expand the ecological and economic positive effects. The improvement policies can provide the feasible policy references for the management and development of the eco-agriculture system in Kongtong District.
(2) The AEP-SD model can reveal specifically the interaction and material flow mechanism among three subsystems of industry, effect and policy; diagnose scientifically the potential shortcomings and defects in the system, providing the basis for making pertinent improvement policies and checking the effectiveness of the improvement policies. The above functions of the AEP-SD model make it dominant in the eco-agriculture system analysis, improvement policies making and decision aiding.
(3) The core modeling thought of the AEP-SD model is that the eco-agriculture system is divided into three parts: industry, effect and policy, and the interactions and material-energy flow mechanism among each part are taken as the logic framework of modeling. This modeling thought is not only effective in the study on the eco-agriculture system, but also has good

application to revealing the internal structures and operate rules in the complex system compounded with ecosystem and other systems such as economy, society or human activity, providing some references and basis for relevant modeling study.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.ecolmodel.2011.12.005.

References

Altieri, M.A., Anderson, M.K., 1986. An ecological basis for the development of alternative agricultural systems for small farmers in the Third World. American Journal of Alternative Agriculture 1 (1), 30–38.
Chang, Y., Hong, F., Lee, M., 2008. A system dynamic based DSS for sustainable coral reef management in Kenting coastal zone, Taiwan. Ecological Modelling 211 (1–2), 153–168.
Chen, G., Jiang, M., Yang, Z., Chen, B., Ji, X., Zhou, J., 2009. Exergetic assessment for ecological economic system: Chinese agriculture. Ecological Modelling 220 (3), 397–410.
Forrester, J.W., 1958. Industrial dynamics: a major breakthrough for decision makers. Harvard Business Review 36 (4), 37–66.
Gale, F.P., 2000. Economic specialization versus ecological diversification: the trade policy implications of taking the ecosystem approach seriously. Ecological Economics 34 (3), 285–292.
Granovskii, M., Dincer, I., Rosen, M.A., 2007. Exergetic life cycle assessment of hydrogen production from renewables. Journal of Power Sources 167 (2), 461–471.
Hau, J.L., Bakshi, B.R., 2004. Expanding exergy analysis to account for ecosystem products and services. Environmental Science & Technology 38 (13), 3768–3777.
Hezri, A.A., Dovers, S.R., 2006. Sustainability indicators, policy and governance: issues for ecological economics. Ecological Economics 60 (1), 86–99.
Hoang, V.N., 2011. Measuring and decomposing changes in agricultural productivity, nitrogen use efficiency and cumulative exergy efficiency: application to OECD agriculture. Ecological Modelling 222 (1), 164–175.
Hoang, V.N., Alauddin, M., 2011. Analysis of agricultural sustainability: a review of exergy methodologies and their application in OECD countries. International Journal of Energy Research 35 (6), 459–476.
Kleinman, P., Pimentel, D., Bryant, R.B., 1995. The ecological sustainability of slash-and-burn agriculture. Agriculture, Ecosystems & Environment 52 (2–3), 235–249.
Larsson, M., Granstedt, A., 2010. Sustainable governance of the agriculture and the Baltic Sea – agricultural reforms, food production and curbed eutrophication. Ecological Economics 69 (10), 1943–1951.
Libralato, S., Torricelli, P., Pranovi, F., 2006. Exergy as ecosystem indicator: an application to the recovery process of marine benthic communities. Ecological Modelling 192 (3–4), 571–585.
Maurer, L., 1989. An ecological approach to agriculture – the Austrian example. Agriculture, Ecosystems & Environment 27 (1–4), 573–578.
Niemeijer, D., 2002. Developing indicators for environmental policy: data-driven and theory-driven approaches examined by example. Environmental Science & Policy 5 (2), 91–103.
Piorr, H.P., 2003. Environmental policy, agri-environmental indicators and landscape indicators. Agriculture, Ecosystems & Environment 98 (1–3), 17–33.
Ropke, I., 2005. Trends in the development of ecological economics from the late 1980 to the early 2000. Ecological Economics 55 (2), 262–290.
Schroll, H., 1994. Energy-flow and ecological sustainability in Danish agriculture. Agriculture, Ecosystems & Environment 51 (3), 301–310.
Sciubba, E., 2003. Extended exergy accounting applied to energy recovery from waste: the concept of total recycling. Energy 28 (13), 1315–1334.
Shi, T., Gill, R., 2005. Developing effective policies for the sustainable development of ecological agriculture in China: the case study of Jinshan County with a systems dynamics model. Ecological Economics 53 (2), 223–246.
Smith, O., Petersen, G., Needelman, B., 1999. Environmental indicators of agroecosystems. Advances in Agronomy, 6975–6997.
Tao, Z., 2010. Scenarios of China's oil consumption per capita (OCPC) using a hybrid Factor Decomposition-System Dynamics (SD) simulation. Energy 35 (1), 168–180.
Wang, Y., Zhang, X., 2001. A dynamic modeling approach to simulating socioeconomic effects on landscape changes. Ecological Modelling 140 (1–2), 141–162.
Yunlong, C., Smit, B., 1994. Sustainability in agriculture: a general review. Agriculture, Ecosystems & Environment 49 (3), 299–307.